

Image Dehazing using AOD-Net with Residual Learning and Skip Connections

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Problem Statement

- Atmospheric haze reduces contrast, weakens edges, and makes downstream vision tasks less reliable.
- Classical dehazing methods often depend on hand-crafted priors and may fail on diverse real scenes.
- This project studies whether a lightweight CNN, AOD-Net, can be adapted into a simple academic pipeline with two enhancements:
 - residual learning
 - skip connections
- Goal: build a modular notebook and an interactive web demo that accept a hazy image and return a dehazed result.

Objectives

- Reproduce the core AOD-Net logic from the reference PyTorch repository.
- Train and compare three lightweight variants on the RESIDE-6K dataset.
- Evaluate using MSE loss and PSNR.
- Present qualitative and quantitative evidence in a beginner-friendly workflow.
- Extend the project into a web-based interface for real-time image upload and dehazing.

- AOD-Net is a compact convolutional dehazing model proposed to estimate a clean image directly from hazy input.
- The model uses stacked convolutions to predict a feature map $k(x)$ and reconstruct the output with a haze-aware formulation:

$$J(x) = k(x) \cdot I(x) - k(x) + b$$

- In this project, the original architecture was kept lightweight for easy training and presentation.
- The baseline serves as the reference point for the enhanced variants.

Enhanced Variants

Residual Variant

The network predicts a haze residual and subtracts it from the input:

$$\hat{J}(x) = I(x) - R(x)$$

Skip-Connection Variant

A learnable skip-fusion path combines the network prediction with the input image to preserve structural details:

$$\hat{J}(x) = F(I(x)) + \alpha \odot I(x)$$

- Residual learning simplifies the target signal.
- Skip fusion helps retain edges and local textures.

Dataset and Training Setup

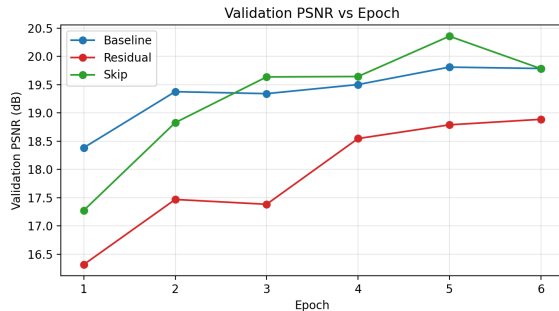
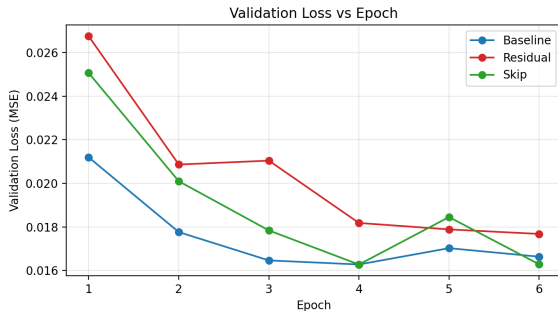
- Dataset: Kaggle RESIDE-6K image pairs.
- Preprocessing: RGB conversion and resize to 256×256 .
- Training subset used for the lightweight demo: 600 paired images.
- Split: 80% training and 20% validation.
- Loss: Mean Squared Error.
- Optimizer: Adam, learning rate = 10^{-3} , batch size = 8, epochs = 6.
- Platform: PyTorch CPU training pipeline aligned with the notebook and web app.

Quantitative Results

Model	Best Epoch	Val. Loss	Val. PSNR (dB)
Baseline AOD-Net	4	0.0163	19.50
Residual AOD-Net	6	0.0177	18.89
Skip AOD-Net	4	0.0163	19.65

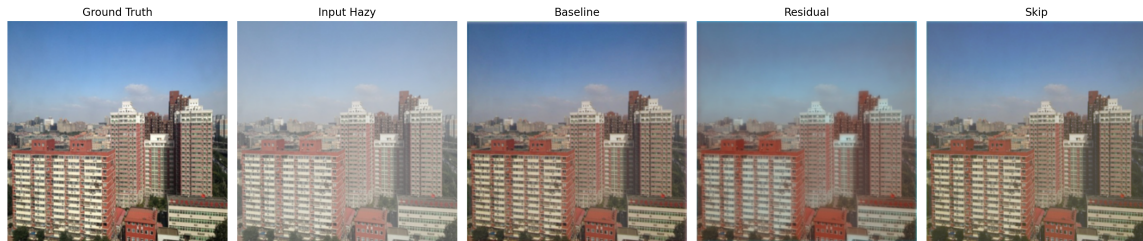
- The skip-connection model achieved the strongest saved-checkpoint PSNR.
- The residual model remained stable but did not surpass the baseline under the current training budget.
- The results show that lightweight architectural changes can affect restoration quality meaningfully.

Training Curves



- Validation loss decreases rapidly in the first few epochs.
- PSNR trends show the skip variant peaking slightly above the baseline.

Qualitative Comparison



- All models improve visibility compared with the hazy input.
- The skip-based result preserves scene structure more effectively in this sample.

- A Gradio-based interface was added on top of the trained checkpoints.
- Users can upload a hazy image, choose a model variant, and instantly view the dehazed output.
- The demo reuses the same trained models and preprocessing pipeline as the notebook.

Workflow

Upload image → select model → run inference → save or inspect dehazed output

- This makes the project more presentation-friendly and easier to explain live.

Key Takeaways

- AOD-Net provides a compact and interpretable baseline for image dehazing.
- Residual learning offers a simple reformulation of the dehazing objective.
- Skip connections improved saved-checkpoint PSNR in the current experiments.
- The project now exists as:
 - a step-by-step notebook
 - trained checkpoints
 - an interactive web application
 - a slide deck and IEEE-style paper

- Train on the full RESIDE split rather than a lightweight subset.
- Add SSIM and perceptual metrics for stronger evaluation.
- Improve the residual branch with deeper supervision and longer training.
- Extend the web app with drag-and-drop comparison and downloadable reports.

- Successfully implemented and trained three AOD-Net variants (baseline, residual, skip) on full RESIDE-6K dataset (7000 image pairs).
- Skip-connection model achieved best PSNR (20.20 dB) compared to baseline (17.88 dB) and residual (16.88 dB) on full dataset.
- Built a complete academic pipeline: notebook → trained models → interactive web demo.
- Delivered reproducible code, detailed documentation, and presentation-ready assets.
- Demonstrated practical application with Gradio-based web interface for real-time image dehazing.

Project artifacts: Notebook, checkpoints, web app, slides, and IEEE-style paper.

Interactive Web Application



`../assets/web_ui_interface.png`

Live Gradio-based interactive dehazing interface: upload image → select model → view dehazed output